

ODD-COMPONENT COMPLEMENTATION: POINTWISE SPLIT CLOCKS AND AN ALL-DEPTH LABELLED CENSUS

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ABSTRACT. On a finite simple labelled graph, synchronously complement the subgraph induced by every connected component of odd order and leave every even component unchanged. We show that the resulting finite dynamics is linearized pointwise by a parity-pruned component/co-component split tree. Its active height is the exact entrance time into the recurrent set, all periods divide two, and the largest transient depth on n vertices is the sharp value $\lfloor (n-1)/2 \rfloor$. We then derive a recursive labelled exponential generating function for every cumulative and exact depth layer. It yields the fixed and recurrent censuses, the number of two-cycles, and the finite-order dynamical zeta function. Gallai decomposition, cographs and cotrees, labelled SET calculus, and connected-graph enumeration are treated as background. The direct-owner search is bounded; no novelty, priority, or external-release claim is made.

1. THE MAP AND THE SUBTRACTION BOUNDARY

All graphs are finite, simple, and labelled. For a graph G , define

$$(1) \quad \Phi(G) = \bigsqcup_{C \in \text{Comp}(G)} \begin{cases} \overline{G[C]}, & |C| \text{ odd}, \\ G[C], & |C| \text{ even}. \end{cases}$$

Thus no edge is added between distinct components, and every odd component is complemented synchronously. We regard K_1 as connected and co-connected. The *depth* of a state is its first entrance time into the recurrent set.

Component/co-component decomposition belongs to classical modular decomposition, beginning with Gallai's work [6]. Cographs, their cotrees, and recognition are likewise classical [4, 5]; modern labelled cotree enumeration appears, for example, in [2, 8]. Labelled SET constructions and connected-graph enumeration are standard [3, 7]. None of those facts, nor the co-connected count used below, receives contribution credit here. Our residual question is temporal: what is the exact clock of the literal parity-triggered map (1), and can every clock layer be counted?

Lemma 1.1 (refinement). *The component partition of $\Phi(G)$ refines that of G . Every even component is fixed forever. If H is odd and connected, then either $\overline{H}$ is connected and $H \leftrightarrow \overline{H}$, or the components of $\overline{H}$ are permanent separate blocks after one step.*

Proof. Equation (1) never creates a cross-component edge. An even block is copied unchanged. On an odd connected block H , one step produces $\overline{H}$. If that graph is connected, the next step complements it back to H . If it is disconnected, no later step can join its components, so they evolve independently. $\square$

2. THE POINTWISE SPLIT CLOCK

The classical component/co-component tree is useful here only after a parity pruning that records which blocks remain dynamically active.

Definition 2.1. For a connected odd graph H , build its *parity-pruned split tree* as follows. If $\overline{H}$ is connected, stop. Otherwise create the connected components of $\overline{H}$ as children; declare every even child a leaf and recurse only on odd children. Let $D(H)$ be the largest number of split nodes on a root-to-leaf branch. Equivalently,

$$(2) \quad D(H) = \begin{cases} 0, & \overline{H} \text{ connected}, \\ 1 + \max\{D(C) : C \in \text{Comp}(\overline{H}), |C| \text{ odd}\}, & \overline{H} \text{ disconnected}, \end{cases}$$

where the maximum of the empty set is 0. For arbitrary G , put $D(G) = \max D(C)$ over its odd components, with empty maximum 0.

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This is not asserted to be a new cotree or modular decomposition. It is a pointwise clock obtained by pruning the familiar split operation according to the update parity.

Theorem 2.2 (exact entrance clock). *For every graph G ,*

$$\text{depth}(G) = D(G).$$

Proof. Even components are recurrent fixed blocks. Distinct odd components evolve independently, so the entrance time of their disjoint union is the maximum of their entrance times. It remains to consider an odd connected H . If $\overline{H}$ is connected, H already lies on a cycle. Otherwise the first step splits H into the components of $\overline{H}$. Its even children freeze, while its odd children require exactly their individual entrance times. Hence the entrance time satisfies precisely (2); induction on $|H|$ finishes the proof. $\square$

Corollary 2.3 (periodic classification). *Every eventual period is 1 or 2. A graph is recurrent exactly when every nontrivial odd component is co-connected. It is fixed exactly when every component is either a singleton or has even order. Every other recurrent graph lies on a genuine two-cycle.*

Proof. Theorem 2.2 reduces every state to leaves of the split tree. Even leaves are fixed. A singleton is fixed by complementation, whereas a nontrivial odd co-connected component alternates with its connected complement. A disjoint union synchronizes these periods. $\square$

3. THE SHARP GLOBAL CLOCK

Theorem 3.1 (sharp maximum depth). *Among all graphs on $n \geq 1$ vertices,*

$$(3) \quad \max_{|V(G)|=n} \text{depth}(G) = \left\lfloor \frac{n-1}{2} \right\rfloor.$$

The value for $n = 0$ is 0.

Proof. Follow an active branch from an odd parent H to an odd child C . The complement $\overline{H}$ has at least two components, and the vertices outside C have even positive total order because both $|H|$ and $|C|$ are odd. Thus $|C| \leq |H| - 2$. Every split consumes at least two vertices along the branch, giving $D(H) \leq (|H| - 1)/2$. Taking the maximum over odd components proves the upper bound for every G .

For sharpness, let $H_1 = K_1$ and recursively set

$$(4) \quad H_{2r+1} = \overline{H_{2r-1} \sqcup K_2}.$$

The complement of H_{2r+1} has the odd child H_{2r-1} and the even child K_2 , so $D(H_{2r+1}) = 1 + D(H_{2r-1}) = r$. This attains (3) for odd order. For even $n = 2r + 2$, the graph $H_{2r+1} \sqcup K_1$ has depth r , proving sharpness there as well. $\square$

The bound is structural, not an extrapolation from the finite census in [section 5](#).

4. ALL-DEPTH LABELLED GENERATING FUNCTIONS

Let c_n be the number of connected labelled graphs on $[n]$, and work throughout with formal exponential generating functions. Put

$$(5) \quad C_e(x) = \sum_{\substack{n \geq 2 \\ n \text{ even}}} c_n \frac{x^n}{n!}.$$

For odd n , let q_n count connected graphs whose complements are also connected, and define

$$(6) \quad q_1 = 1, \quad q_n = 2c_n - 2\binom{n}{2} \quad (n \geq 3 \text{ odd}), \quad Q(x) = \sum_{n \text{ odd}} q_n \frac{x^n}{n!}.$$

Indeed, a graph and its complement cannot both be disconnected when $n \geq 2$, so (6) is inclusion-exclusion; this identity is zero-credit background. For a series $A(x)$, write $\text{Odd } A = (A(x) - A(-x))/2$.

Theorem 4.1 (complete depth recurrence). *Let $O_t(x)$ be the EGF of connected odd graphs of depth at most t , and let $F_t(x)$ be the EGF of all graphs of depth at most t . Then*

$$(7) \quad O_0(x) = Q(x),$$

$$(8) \quad O_t(x) = Q(x) + \text{Odd}(\exp(C_e(x) + O_{t-1}(x)) - 1 - C_e(x) - O_{t-1}(x)), \quad t \geq 1,$$

$$(9) \quad F_t(x) = \exp(C_e(x) + O_t(x)).$$

Consequently, for $t \geq 1$, the number of labelled n -vertex graphs of exact depth t is

$$(10) \quad n![x^n](F_t(x) - F_{t-1}(x)),$$

while $n![x^n]F_0(x)$ is the recurrent count.

Proof. Depth-zero connected odd graphs are precisely the co-connected ones, giving (7). Consider a positive-depth connected odd graph H . Complementation bijects it with a disconnected labelled graph whose connected components have odd total order. Its even components are unrestricted connected graphs, counted by C_e ; by (2), every odd component has depth at most $t-1$, counted by O_{t-1} . The labelled SET construction gives the exponential in (8); subtracting $1 + C_e + O_{t-1}$ forces at least two components, and odd extraction forces odd total order. Adding the co-connected base class Q proves (8).

An arbitrary graph of depth at most t is a SET of unrestricted even connected components and odd connected components counted by O_t . The exponential formula gives (9), and subtraction of successive cumulative classes gives (10). $\square$

Proposition 4.2 (fixed census). *The EGF for fixed graphs is*

$$(11) \quad F_{\text{fix}}(x) = \exp(x + C_e(x)).$$

Proof. By corollary 2.3, the allowed connected components are singletons and arbitrary even connected graphs. Apply the labelled SET construction. $\square$

5. RECURRENT AND ZETA CENSUS

Write

$$f_n = n![x^n]F_{\text{fix}}(x), \quad r_n = n![x^n]F_0(x).$$

Thus f_n and r_n are respectively the fixed and recurrent state counts on the finite state space of labelled graphs on $[n]$.

Corollary 5.1 (two-cycles and zeta). *The number of genuine two-cycles is $(r_n - f_n)/2$. Moreover,*

$$(12) \quad \# \text{Fix}(\Phi^k) = \begin{cases} f_n, & k \text{ odd}, \\ r_n, & k \text{ even}, \end{cases} \quad \zeta_n(z) = \exp\left(\sum_{k \geq 1} \frac{\# \text{Fix}(\Phi^k)}{k} z^k\right) = (1-z)^{-f_n} (1-z^2)^{-(r_n-f_n)/2}.$$

Proof. Corollary 2.3 partitions the recurrent set into f_n one-cycles and $(r_n - f_n)/2$ two-cycles. Transient states are fixed by no positive iterate. Hence the displayed fixed-iterate counts hold. Splitting the defining Artin–Mazur series [1] into odd and even powers and applying the logarithmic series gives (12). $\square$

For orientation, exact coefficients and exhaustive orbits give:

n	f_n	r_n	two-cycles	maximum depth
0	1	1	0	0
1	1	1	0	0
2	2	2	0	0
3	4	4	0	1
4	48	48	0	1
5	216	648	216	2
6	27,920	30,512	1,296	2

The paper-local verifier exhausts every labelled graph through $n = 6$, checks the orbit periods and depth histograms, and independently assembles the coefficients of (8)–(9). It executes 67,758 exact assertions. These computations test definitions and translations; the all-order conclusions rest on the proofs above.

6. SCOPE AND CONCLUSION

The parity-pruned split tree simultaneously exposes the pointwise and enumerative structure of odd-component complementation. It gives an exact entrance clock, the sharp maximum $\lfloor (n-1)/2 \rfloor$, and a recursive EGF for every temporal layer. The recurrent and fixed specializations then produce the two-cycle and zeta censuses without further orbit analysis.

The contribution boundary is deliberately narrow. Gallai decomposition, cographs and cotrees, connected labelled graph counts, and labelled species operators remain zero-credit background. The direct-owner search was bounded rather than exhaustive, and the finite verifier stops at order six. Accordingly, this manuscript makes no novelty or priority assertion and is held from external release pending a broader owner audit.

REFERENCES

- [1] Michael Artin and Barry Mazur. On periodic points. *Annals of Mathematics*, 81(1):82–99, 1965. doi: 10.2307/1970384.
- [2] Frédérique Bassino, Mathilde Bouvel, Valentin Féray, Lucas Gerin, Mickaël Maazoun, and Adeline Pierrot. Random cographs: Brownian graphon limit and asymptotic degree distribution. *Random Structures & Algorithms*, 60(2):166–200, 2022. doi: 10.1002/rsa.21033.
- [3] François Bergeron, Gilbert Labelle, and Pierre Leroux. *Combinatorial Species and Tree-like Structures*, volume 67 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, 1998. doi: 10.1017/CBO9781107325913.
- [4] Derek G. Corneil, Helmut Lerchs, and Lorna Stewart Burlingham. Complement reducible graphs. *Discrete Applied Mathematics*, 3(3):163–174, 1981. doi: 10.1016/0166-218X(81)90013-5.
- [5] Derek G. Corneil, Yehoshua Perl, and Lorna K. Stewart. A linear recognition algorithm for cographs. *SIAM Journal on Computing*, 14(4):926–934, 1985. doi: 10.1137/0214065.
- [6] Tibor Gallai. Transitiv orientierbare Graphen. *Acta Mathematica Academiae Scientiarum Hungaricae*, 18:25–66, 1967. doi: 10.1007/BF02020961.
- [7] Frank Harary and Edgar M. Palmer. *Graphical Enumeration*. Academic Press, 1973.
- [8] Benedikt Stuffer. Graphon convergence of random cographs. *Random Structures & Algorithms*, 59(3):464–491, 2021. doi: 10.1002/rsa.21002.